

20.Future Sales Prediction using Machine Learning

Aim

To implement a **Future Sales Prediction Model** using a suitable machine learning algorithm in Python and predict future sales based on previous sales data.

Algorithm

1. Import the required libraries.
2. Create a dataset containing previous sales and corresponding time periods.
3. Separate the input and target variables.
4. Split the dataset into training and testing data.
5. Create a **Linear Regression** model.
6. Train the model using the training data.
7. Predict sales for the test data.
8. Calculate the model performance using R^2 score.
9. Predict the sales for a future period.

Program:

```
from sklearn.linear_model import LinearRegression

from sklearn.model_selection import train_test_split

from sklearn.metrics import r2_score
```

```
X = [

    [1], [2], [3], [4], [5],

    [6], [7], [8], [9], [10]

]
```

```
y = [

    120, 150, 180, 210, 240,

    270, 300, 330, 360, 390

]
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
score = r2_score(y_test, y_pred)
```

```
print("Actual Sales:", y_test)
```

```
print("Predicted Sales:", y_pred)
```

```
print("R2 Score:", score)
```

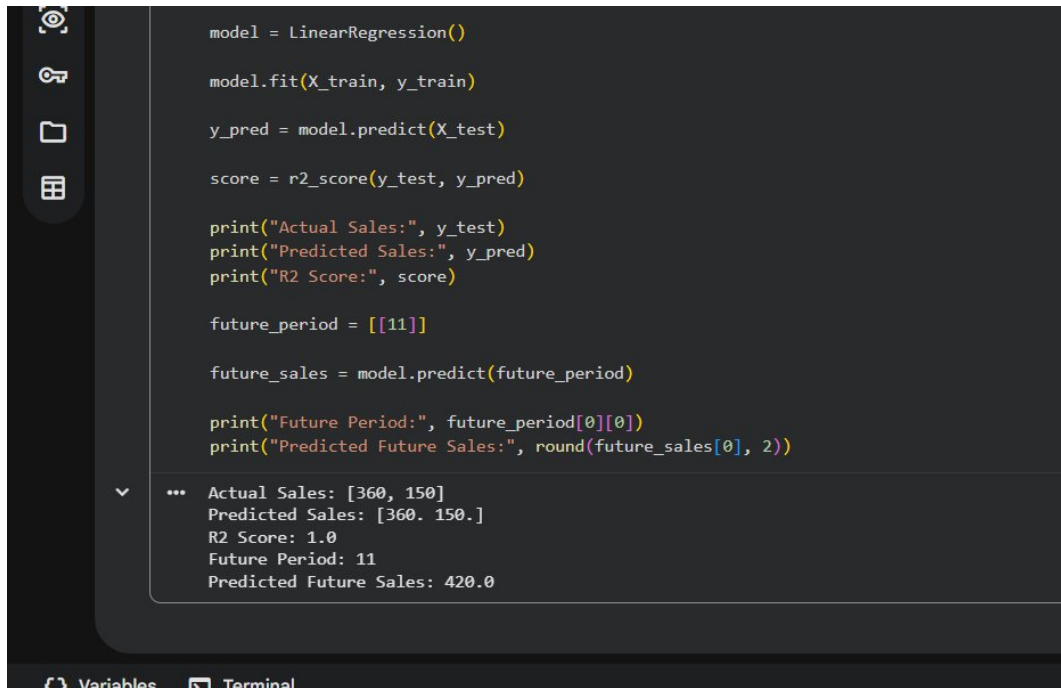
```
future_period = [[11]]
```

```
future_sales = model.predict(future_period)
```

```
print("Future Period:", future_period[0][0])
```

```
print("Predicted Future Sales:", round(future_sales[0], 2))
```

Output:



```
model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

score = r2_score(y_test, y_pred)

print("Actual Sales:", y_test)
print("Predicted Sales:", y_pred)
print("R2 Score:", score)

future_period = [[11]]

future_sales = model.predict(future_period)

print("Future Period:", future_period[0][0])
print("Predicted Future Sales:", round(future_sales[0], 2))
```

... Actual Sales: [360, 150]
Predicted Sales: [360. 150.]
R2 Score: 1.0
Future Period: 11
Predicted Future Sales: 420.0

Variables Terminal

Result Statement

Thus, the **Future Sales Prediction** model was successfully implemented using **Linear Regression**, and the sales for a future period were successfully predicted.